A069 v0.5.0 — Blind 3D Trefoil GP Dynamics and Independent Mode Controls

SST-Workbench / development falsifier, external framework v1.0.9

19 October 2026

Abstract

We extend the prior 1D Hasimoto and exact Euler reference calculations with actual three-dimensional, volume-resolved, time-dependent Gross–Pitaevskii simulations of four historical trefoil configurations, together with independent bulk Bogoliubov dispersion and three-dimensional tangent/BdG propagation checks. An oriented spanning two-chain supplies a globally winding, regularized complex vortex field rather than an invalid nearest-centerline phase ansatz. The current evolution is an *initial-value demonstration*, not a certified equilibrated knotted particle, a solution of incompressible Euler, or a measured internal clock. A stable gauge- and frame-independent trefoil eigenfrequency remains `UNRESOLVED`; frozen physical reveal is prohibited.

1 Identity and epistemic status

A069 (provisional), `v0.5.0`, with an external version pin to `SST_Falsifier_Framework_v1.0.9`. This report can be published by the independent shared framework, whose original installed release is the certification authority. Local reconstruction of framework-module snapshots is only a development smoke test. Previous v0.4.0 outputs are neither replaced nor promoted.

2 Research question

Can true three-dimensional knotted GP dynamics yield a temporally stable, *intrinsic*, model-derived oscillation frequency from knot initial conditions alone? We distinguish: (a) quantized vortex *winding*, (b) numerical 3D time evolution, (c) a genuine reduced linear eigenmode of a stationary or relative-periodic knot, and (d) a further *physical* Swirl Clock. These are four distinct claims.

3 Null hypothesis, alternatives, and blindness

H_0: no stable, gauge/frame-independent, convergent knot eigenmode with cross-provider held-out predicted rate is demonstrated. H_1: the uncalibrated rate from a qualified 3D knot eigenmode predicts independent held-out data. Both remain `UNRESOLVED`. Analytic mode references can pass without promoting either physical hypothesis. Source labels are keyed to opaque case IDs during the official blind run; the historical seeds and their mirrors are correlated test inputs rather than blind physical observations.

4 Dynamical assumptions and excluded hypotheses

Two mutually non-identical theories are retained: classical incompressible Euler in the inherited D1/D2A/D2H reference cases, and the *compressible* GP order parameter for the present 3D knotted initial-value test. GP allows density excitations and reconnection. An arbitrary knot geometry is not a stationary GP solution. No Compton mass, Lorentz factor, delay-mode frequency, observed atomic-clock frequency, golden-ratio constraint, or older visual KnotPlot relaxation is passed to the solver. The original A and B curves are parametrically verified and not assumed to be independent physical providers.

5 Units and source variables

The new calculation uses dimensionless GP units: $\mathbf{x}$ is position, T is evolution time, ξ is an explicitly declared preparatory core length, Ψ is a dimensionless complex field, and L_b is periodic domain length. No conversion to SI

seconds or to SST particle masses is inferred. The source has $N = 1200$ fixed points per curve. The 3D grid has N_g^3 cells, where N_g is independent of filament discretization.

6 Equations and non-circular predictions

6.1 Finite-volume, fully 3D nonlinear Schrodinger model

We integrate the normalized Hamiltonian GP equation,

$$i\partial_T \Psi = -\frac{1}{2}\nabla^2\Psi + (|\Psi|^2 - 1)\Psi, \qquad \mathbf{x} \in [-L_b/2, L_b/2)^3. \tag{1}$$

The modulus of the initialized field vanishes in a finite-width tube about the closed seed curve K. The *phase* is constructed instead from the oriented solid angle $\Omega_\Sigma(\mathbf{x})$ of a spanning 2-chain Σ whose boundary is K:

$$\Psi_{\rm raw}(\mathbf{x}, 0) = \tanh\left(\frac{d(\mathbf{x}, K)}{\sqrt{2}\xi}\right) \exp\left[\frac{i\Omega_\Sigma(\mathbf{x})}{2}\right]. \tag{2}$$

Because Ω_Σ changes by 4π across a spanning sheet, $\exp(i\Omega_\Sigma/2)$ remains single-valued and has local 2π vortex circulation. A smooth real window $\chi(\mathbf{x})$ supported outside the vortex region interpolates the field to the uniform condensate at the periodic faces:

$$\Psi_0 = 1 + \chi(\Psi_{\rm raw} - 1). \tag{3}$$

The taper is *preparation-dependent*: the resulting Ψ_0 is not assumed to be a relaxed stationary GP knot. We require exact periodic-face agreement and nonzero plaquette phase-winding counts.

Split-step spectral Strang integration advances the full N_g^3 complex array in absolute dimensionless time T; mass M, GP energy E, and vortical plaquettes are monitored:

$$M = \int |\Psi|^2 d^3x, \qquad E = \int \left(\frac{1}{2}|\nabla\Psi|^2 + \frac{1}{2}(|\Psi|^2 - 1)^2\right) d^3x. \tag{4}$$

Plaquette counts are coarse, mesh-dependent indicators, *not* a continuous topological linking invariant. A count change alone does not demonstrate a specific reconnection event.

6.2 Independent bulk Bogoliubov-mode reference

Linearization around the exact uniform state $\Psi = 1$ predicts without numerical frequency fitting:

$$\boxed{\omega_B(k) = \sqrt{\varepsilon_k(\varepsilon_k + 2)}, \quad \varepsilon_k = k^2/2.} \tag{5}$$

The independent nonlinear GP integrator evolves a small plane-density perturbation. Its observed oscillation frequency is extracted from a numerical FFT peak and fitted from that peak, *not* searched inside a tolerance window centered on the analytic prediction. A minimum of four complete temporal cycles is mandatory before declaring this reference control PASS.

6.3 Three-dimensional time-dependent BdG tangent response

A second and independent perturbation experiment checks the time-dependent real linearization

$$i\partial_T \delta\Psi = -\frac{1}{2}\nabla^2\delta\Psi + (2|\Psi|^2 - 1)\delta\Psi + \Psi^2\delta\Psi^*, \tag{6}$$

by differentiating the nonlinear Strang map (Fréchet derivative) and comparing with finite differences of two separately evolved nonlinear 3D fields. This tests a *time-dependent tangent propagator*; it does not establish an intrinsic knot eigenvalue without a qualified stationary/relative-periodic base.

6.4 Required internal-mode interpretation

A candidate shape observable is built from the eigenvalues of a translationally centered, rotation-invariant covariance of the depleted condensate density. Its global $U(1)$ phase is irrelevant. An apparent spectral peak must be independently predicted on unseen later time series, persist over ≥ 4 complete cycles, converge under N_g, ΔT, ξ, and domain changes, and survive removal of collective translation, rotation, and global-phase modes. None of these conditions can be inferred from the bulk Bogoliubov reference.

Machine-readable science contract

Research question Can 3D GP knot evolution and an independently predicted co-moving intrinsic knot eigenfrequency be resolved for historical trefoils without any clock or Lorentz input?

Objective Run genuine 3D GP knot initial-value dynamics and independent Bogoliubov/tangent controls; certify knot eigenfrequency only if co-moving stationary/periodic, shape-invariant frequency and independent holdouts pass.

H_0 No convergent, persistent internal shape-mode and target-free 3D physical clock-frequency closure is certified by these inputs.

H_1 After qualified 3D vortex evolution and symmetry reduction, a preregistered intrinsic mode predicts independent held-out phase time series within fixed tolerances without per-geometry refitting.

Registered equations

E1: Kelvin

$$\Gamma = \oint_{C(T)} \mathbf{u} \cdot d\ell = 2\pi R(T)^2 \Omega(T) \tag{7}$$

Dimensional check: m^2/s on both sides

E2: Exact affine 3D Euler

$$\mathbf{u} = A(T)\mathbf{x}, \quad A = \begin{pmatrix} -a/2 & -\Omega & 0 \\ \Omega & -a/2 & 0 \\ 0 & 0 & a \end{pmatrix}, \quad \dot{\Omega} = a\Omega \tag{8}$$

Dimensional check: 1/s times m gives m/s

E3: First-principles analytic control frequency (not knot clock)

$$\Omega(T) = \Omega_0 e^{aT} = \Gamma/[2\pi R(T)^2] \tag{9}$$

Dimensional check: 1/s

E4: Tracer phase

$$\theta(T) = \int_0^T \Omega(s) ds = \Omega_0 \,\mathrm{expm1}(aT)/a \tag{10}$$

Dimensional check: dimensionless

E5: Rotating reference frame identity

$$\boldsymbol{\omega}_{abs} = \boldsymbol{\omega}_{rel} + 2\boldsymbol{\Omega}_{frame} \tag{11}$$

Dimensional check: 1/s

E6: Euler pressure Hessian analytic control

$$\nabla\nabla p = -\rho \,\mathrm{sym}(\dot{A} + A^2) \tag{12}$$

Dimensional check: kg m^-3 s^-2

E7: Exact smooth finite-core columnar Euler flow

$$u_\theta(r) = Cre^{-(r/a)^2}, \quad \omega_z(r) = 2C(1 - r^2/a^2)e^{-(r/a)^2} \tag{13}$$

Dimensional check: [C]=1/s, [u_theta]=m/s, [omega_z]=1/s

E8: Stationary Euler pressure balance

$$\partial_r p = \rho r C^2 e^{-2r^2/a^2} \tag{14}$$

Dimensional check: Pa/m = (kg/m^3)(m)(s^-2)

E9: Independent material tracer phase check

$$\dot{\mathbf{x}} = \mathbf{u}(\mathbf{x}), \quad \theta_{\text{pred}}(T) = Ce^{-(r_0/a)^2}T \tag{15}$$

Dimensional check: dimensionless phase from 1/s times s

E15: Hill exact vorticity and propagation identity

$$\omega_\phi = A_H s \ (r < a), \quad U_H = \frac{2}{15}a^2 A_H = \frac{\Gamma_H}{5a} \tag{16}$$

Dimensional check: A_H [m^-1 s^-1] * s [m] => [s^-1], a^2*A_H => [m s^-1]

E16: Stationary 2D GP m=1 vortex radial boundary problem

$$-\frac{1}{2}\left(f'' + \frac{f'}{r} - \frac{f}{r^2}\right) + (f^2 - 1)f = 0 \tag{17}$$

Dimensional check: All terms dimensionless in healing-length and inverse-chemical-potential time units.

E17: Energy impulse velocity law, with core-size derivative explicitly constrained

$$E = \frac{1}{2}\rho\Gamma^2 R(\ln(8R/a) - \alpha), \quad P = \rho\Gamma\pi R^2, \quad U = \frac{dE}{dP} \tag{18}$$

Dimensional check: E [kg m^2 s^-2]; P [kg m s^-1]; dE/dP [m s^-1].

E18: E011 static gate with immutable hash and legacy family exclusion

$$\text{accept}_{\text{static}} = \text{SHA256}_{\text{pinned}} \wedge \text{PRIMARY} \wedge \text{capabilities} \wedge \neg\text{legacy} \tag{19}$$

Dimensional check: Logical equality/selection; no physical frequency dimensions.

E19: Frenet Hasimoto mapping

$$\psi_H(s) = \kappa(s)\exp\left(i\int_0^s \tau(\sigma)\,d\sigma\right), \quad \Phi_H = \oint \tau\,ds \tag{20}$$

Dimensional check: dimensionless geometric normalization, inverse length amplitude

E20: Twisted periodic gauge and focusing LIA NLS

$$\psi_H = e^{i\delta s}q, \quad \delta = \Phi_H/L, \quad iq_T + \beta_{LIA}\{(\partial_s + i\delta)^2 q + \tfrac{1}{2}|q|^2 q\} = 0 \tag{21}$$

Dimensional check: Dimensionless s and T, beta_LIA sets nondimensional time

E21: NLS mass and covariant Hamiltonian

$$M = \int_0^L |q|^2 ds, \quad H = \int_0^L \left(|(\partial_s + i\delta)q|^2 - \tfrac{1}{4}|q|^4 \right) ds \tag{22}$$

Dimensional check: dimensionless numerical invariants with frozen length scale

E22: Relative equilibrium residual under finite-core Biot Savart

$$\mathcal{R}_{RE} = \frac{\min_{U,\Omega,\lambda} \|v_{BS} - U - \Omega \times X - \lambda X_s\|_2}{\max(\|v_{BS}\|_2, v_{floor})} \tag{23}$$

Dimensional check: dimensionless ratio; not evaluated without qualified dynamic geometry

E26: 3D GP

$$i\partial_T \Psi = -\frac{1}{2}\nabla^2 \Psi + (|\Psi|^2 - 1)\Psi \tag{24}$$

Dimensional check: dimensionless

E27: Circulation-winding init

$$\Psi_0 = (1 - \chi) + \chi \tanh(d/(\sqrt{2}\xi)) \exp[i\Omega_\Sigma/2] \tag{25}$$

Dimensional check: dimensionless

E28: Bogoliubov reference dispersion

$$\omega_B(k) = \sqrt{(k^2/2)(k^2/2 + 2)} \tag{26}$$

Dimensional check: dimensionless frequency

E29: Frechet linearization

$$i\partial_T \delta\Psi = -\frac{1}{2}\nabla^2 \delta\Psi + (2|\Psi|^2 - 1)\delta\Psi + \Psi^2 \delta\Psi^* \tag{27}$$

Dimensional check: dimensionless

Registered code-to-mathematics steps

1. **S0** [G1; equations E1]: Audit provenance and refuse old relaxed KnotPlot/VortexLab data
2. **S1** [G2; equations E2, E6]: Construct divergence-free affine Euler reference and exact pressure Hessian
3. **S2** [G2; equations E1, E3, E5]: Check Kelvin loop transport and rotating reference decomposition
4. **S3** [G3; equations E2, E4]: Integrate Cartesian material tracer with RK4 and midpoint, calculate phase via atan2 unwrap
5. **S4** [G4; equations E2]: Require true 3D finite-core dynamic certification before PKLSA prediction and reveal
6. **S12** [G14; equations E2]: Load previous D1 raw JSON and compare SHA-256 with fixed signed baseline
7. **S13** [G12; equations E7, E8]: Use smooth columnar Euler exact field to calculate pressure gradient and vorticity with independent Cartesian finite differences
8. **S14** [G13; equations E7, E9]: Evolve Cartesian Lagrangian tracers with midpoint/RK4 at 160/320/640 steps and compare measured azimuth with analytic Euler prediction without introducing any clock factor
9. **S15** [G15; equations E15]: Compute exact interior/exterior Hill flow, check Euler residual and boundary continuity

10. **S16** [G16; equations E16]: Solve GP radial BVP at two radial grids, differentiate with separate stencil, audit phase-gauge null
11. **S17** [G17; equations E17]: Differentiate thin-ring E(R) numerically and compare v=dE/dP to beta laws under frozen core kinematics
12. **S18** [G18; equations E18]: SHA256 lock E011 atlas ZIP and select provider anchors by capability; reject KnotPlot and mark dynamics unresolved
13. **S20** [G20; equations E19]: Hash and parse two original 2015 sources and generate separately mirror-conjugated controls; do not treat as independent physics providers
14. **S21** [G21; equations E19, E20]: Compute analytic derivatives, resample arclength, extract signed total torsion, and construct holonomy-covariant Hasimoto initial data
15. **S22** [G22; equations E20, E21]: Evolve focusing 1D Hasimoto NLS with split-step FFT and independent RK4, report invariant budgets and step refinement
16. **S23** [G23; equations E19, E20]: Test exact parity mirror with reversed signed evolution coefficient and changed torsion holonomy; reject naive same-coefficient parity criterion
17. **S24** [G24; equations E22]: Block physical 3D GP/Euler and intrinsic clock conclusions without validated 3D order-parameter phase, finite-core pressure and provider holdouts
18. **S26** [G26; equations E26, E27]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)
19. **S27** [G27; equations E28]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)
20. **S28** [G28; equations E29]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)
21. **S29** [G29; equations E29]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)
22. **S30** [G30; equations E26]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)
23. **S31** [G31; equations E26]: New 3D dynamical science gate (see experiment/knotted_gp3d.py)

7 Source-locked algorithm

(1) Validate the original unrelaxed 2015 source SHA-256 values and analytical $N = 1200$ geometry; (2) compute A, mirror(A), B, mirror(B), with two stated grid resolutions for selected non-mirror cases; (3) initialize solid-angle phase and smooth periodic amplitude; (4) integrate all three spatial dimensions; (5) evaluate mass, energy, phase winding and neutral shape diagnostics; (6) validate an independent bulk Bogoliubov reference over multiple cycles; (7) validate real GP tangent dynamics by independent nonlinear perturbation; (8) declare knot eigenfrequency `UNRESOLVED` until full topological, stability and holdout gates close. A SHA-256 pinned JSON configuration specifies the exact parameters and is itself pinned in the preregistered science contract.

8 Provenance and source independence

Two closed centerlines originate from one 2015 research notebook, not two independent physical experiments. Exact reflected curves are parity controls. E011 static atlas metadata is not `DYNAMICS_READY`; older KnotPlot relaxation remains excluded from physical evidence. The prepared 3D GP initial state is generated *from those supplied centerlines*; no provider-certified 3D time series was used. Analytic bulk modes and tangent tests are software/model checks, not separate physical providers.

9 Blind gate interpretation

Gate	Description	Allowed classification
G0–G3, G12–G23	Inherited Euler, GP radial and Hasimoto 1D model controls	Reference PASS/FAIL
G24–G25	Prior full physical knot eigenclock and dynamics certification	UNRESOLVED

G26	Actual 3D GP knot initial-value time integration, phase defects, norm and energy	Model PASS/FAIL
G27	Independent bulk Bogoliubov frequency with ≥ 4 resolved cycles	Reference PASS/FAIL
G28	Time-dependent 3D GP tangent/BdG versus independent finite differences	Model PASS/FAIL
G29	True gauge/frame-independent stable trefoil eigenfrequency	UNRESOLVED
G30	Full knot-topology, numerical convergence, Euler/GP bridge	UNRESOLVED
G31	Independent physical clock-source/A061 downstream closure	UNRESOLVED
G4–G11	Inherited physical certification and provenance gates	UNRESOLVED or disabled

Every explicit model PASS is prevented from promoting a physical claim. An incomplete input fails closed. Physical reveal is blocked when G24, G25, G29, G30 or G31 remains unresolved.

10 Reproducible numerical campaign

For this development preflight the preregistered grid sizes are 48^3 and 64^3, $L_b = 10$, $\xi = 0.42$, $\Delta T = 0.006$, and 160 steps (dimensionless $T_{\max} = 0.96$). A run on each original geometry and its mirror uses 48^3; an independent spatial refinement on each original geometry uses 64^3. Mass relative drift must remain below 10^{-9}, energy relative drift below 10^{-3} and nonzero initial vortex winding must be detected. These are conservative *numerical preflight* limits: they do not certify convergence, a stable knotted structure or a clock. The long-duration production profile must use new, immutable run IDs and separately extended time, grid, core and domain ladders.

11 Backend authority

Python NumPy/SciPy FP64 with Numba-accelerated phase initialization is the implementation authority for the initial-value demonstration. No C++/SYCL backend parity, pressure-Poisson Euler evolution, or Arc A770 certification is claimed. GP is an order-parameter model, not the incompressible Euler substrate.

12 Blindness

New 32-byte nonces and source keys belong only to the PRIVATE package. Outcome files use opaque HMAC IDs; the correspondence to public seed labels is held separately. Historical source identity is known, so this cannot masquerade as a blinded experimental source; a future genuine held-out provider remains necessary. Reveal requires an official qualified framework run and all prerequisite physical gates.

13 Statistical epistemics

Replaying the same centerline at two grids and reflecting its chirality do not create independent samples. No frequentist significance, independent replication, stable mode detection or physical posterior probability is claimed. A bulk reference eigenfrequency does not count as a knotted internal mode.

14 Automatically generated framework results

Automatic run provenance

Field	Recorded value
Run ID	20261019T110408419035Z-427d2e83
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	3bd175f1e2430222d913ea8aef6143003bc6eee53fa8749e296fe8d170416e86
Portable protocol diagnostic SHA-256	9c6489be7addbff1d81e598d75a33fdef908abaa55abe7b86a404c340c45d491
Framework schema	SST-FALSIFIER-PROTOCOL-2

Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	f5c096ca236446efde8e8ce4a372f301dd91d91c8c4a47b7cd47b898c6e54c55
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is official framework protocol integrity verified?	v1.0.9 protocol bundle frozen; external framework-release identity requires qualified upstream host audit
G1	PASS	Were historical KnotPlot and unqualified visual relaxations excluded from scientific D1 inputs?	No historical KnotPlot or VortexLab geometry admitted as physics
G2	PASS	Does exact affine 3D Euler control satisfy divergence, Euler, Kelvin and vortex-stretching identities?	Inherited affine Euler control
G3	PASS	Does a Lagrangian Cartesian tracer yield the independently predicted control angular phase?	Inherited tracer phase control only
G12	PASS	Does the smooth finite-core columnar exact Euler control satisfy pressure, curl, incompressibility and phase tests?	Inherited columnar exact Euler only
G13	PASS	Do two time-integrators and the 160/320/640 refinement ladder reproduce the analytic columnar phase without clock fit?	Inherited time-integrator control
G14	PASS	Does the inherited D1 dataset match the preregistered original SHA-256 and independently regenerated D1 reference?	Inherited D1 exact SHA match
G15	PASS	Does exact Hill spherical 3D Euler satisfy pressure, interior curl, boundary continuity and Kelvin speed identity?	Hill sphere exact 3D Euler; not PKLSA or clock
G16	PASS	Does the dimensionless GP radial vortex solve its boundary problem with a converged residual and gauge-null check?	Dimensionless straight GP vortex, gauge phase not a physical tick
G17	PASS	Does dE/dP reproduce the thin-ring constant-core/volume velocity identities without thick-core extrapolation?	Thin-ring analytic identities only; no fat-core or physical time prediction
G18	PASS	Does E011 archive pass immutable SHA and fail-closed static provider selection without admitting KnotPlot relaxation?	E011 metadata verified; KnotPlot excluded, STATIC is NOT DYNAMICS
G19	UNRESOLVED	Do the published analytic solutions yield a qualified stable intrinsic vortex eigenmode with zero frame/gauge ambiguity?	No gauge-invariant, converged, stable intrinsic 3D vortex eigenclock
G4	NOT_APPLICABLE	Is a true finite-core 3D Euler vortex trajectory dynamically qualified?	Minimal reference profile disabled; No qualified 3D finite-core PKLSA Euler trajectory; Hill not a particle knot
G5	NOT_APPLICABLE	Is there a stable internal mode phase independent of mesh gauge and background rotation?	Minimal reference profile disabled; No stable gauge-invariant internal 3D knot eigenmode
G6	NOT_APPLICABLE	Are E010/E011 origins SHA-verified and dynamically qualified rather than only STATIC_READY?	Minimal reference profile disabled; Static atlas checked, dynamic upstream source and physical qualification absent
G7	NOT_APPLICABLE	Does an independently precommitted rate prediction match the held-out mode phase?	Minimal reference profile disabled; No physical holdout frequency precommitment

G8	UNRESOLVED	Do temporal/spatial/core and pressure-Poisson convergence, energy and helicity checks pass?	3D evolving pressure Poisson, time/space/core convergence not performed
G9	UNRESOLVED	Does provider-independent PKLSA holdout confirm signed rate and uncertainty without refitting?	Independent physics providers and phase time series absent
G10	UNRESOLVED	Has actual strict C++ FP64 certification executed for the physical numerical kernel?	Strict physical native C++ kernel certification unperformed
G11	UNRESOLVED	Is downstream A061 transfer permitted by all upstream science qualification gates?	No physical clock transfer to A061
G20	PASS	Do both historical trefoil sources pass exact byte SHA and no legacy relaxation qualification?	2015 exact bytes and shape checked; original unrelaxed analytic seeds only
G21	PASS	Do analytic curvature and torsion produce converged holonomy-covariant initial Hasimoto fields?	Analytic Frenet derivatives, total torsion and holonomy-covariant init; no physical units
G22	PASS	Does actual 1D Hasimoto-NLS evolution satisfy frozen invariants and independent integrator parity?	Genuine time-evolved 1D LIA NLS + separate 3D GP plane wave; NOT 3D trefoil
G23	PASS	Does reversal of signed LIA evolution under exact mirror obey the correct parity conjugacy?	Mirror requires beta_LIA sign reversal; pure shape parity control, correlated samples
G24	UNRESOLVED	Is a physical 3D knot with qualified stable intrinsic clock mode independently predicted and held-out?	No physical finite-core 3D trefoil evolution, reduced Floquet mode or independently held-out clock target
G25	UNRESOLVED	Can 3D Gross-Pitaevskii trefoil dynamics pass a validated phase, finite-core and reconnection/energy audit?	3D trefoil GP winding and initial-value evolution now executed in G26, but no certified topology, lifetime, core/grid convergence, independent full dynamics provider or stable eigenclock
G26	PASS	Does 3D GP knot initial-value evolution preserve discrete mass/energy numerically and contain initial quantized winding?	Genuine GP x-y-z finite-volume FFT evolution; knots are initial phase defects; dynamics/cross-grid knot topology not certified
G27	PASS	Can an independently predicted bulk Bogoliubov frequency match an independent GP time series?	Independent bulk GP Bogoliubov dispersion: genuine reference eigenmode, NOT a trefoil mode
G28	PASS	Does the exact time-dependent three-dimensional BdG tangent agree with independent finite differences?	Time-dependent 3D GP tangent/BdG response validated against finite-difference independent evolution; no stationary base
G29	UNRESOLVED	Has a stable, gauge- and frame-independent knot-intrinsic eigenfrequency been derived and held out?	No stationary/relative-periodic knot base, reduced Floquet spectrum or >=4 converged frame-invariant cycles; no internal knot eigenfrequency
G30	UNRESOLVED	Is full 3D knot topology preserved and dynamics/core/time/space converged under qualified GP/Euler sources?	3D knot topology tracking across time and spatial/time/core convergence not certified; GP compressible, not Euler
G31	UNRESOLVED	Can independently verified particle-time frequencies be transferred to A061 without parameter fitting?	No independent physical source, experimental clock, or A061-ready proper-time closure

Backend manifest


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{
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Source manifest


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      "source_id": "S1",
      "version": "v0.3.0"
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      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "same_author_historical_2015",
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ourier_2015_p32_N1200_arclength.xyz",
      "provenance_family": "same_2015_notebook",
      "registry_hint": {
        "catalog_id": "A069",
        "hits": []
      },
      "role": "static_geometry_control_only",
```

```
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      "source_id": "S2015_Fourier_2015_p32",
      "version": "2015-original-parametric"
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    {
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      "evidence_class": "synthetic_control",
      "exists": true,
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ers\\A069_3D_SwirlClock_Emergence\\A069-v0.5.0\\sources\\HISTORICAL_2015\\B_T
orus_2015_p66_N1200_arclength.xyz",
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      "version": "2015-original-parametric"
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  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Conclusions and next falsification

A genuine 3D complex GP knot field now evolves with finite spatial resolution. This provides an *operational numerical foundation* for investigating whether the original knot survives. The current run is too short for qualified long-term knot topology or an internal eigenfrequency, and the initial field is nonstationary. The central A069 Swirl Clock hypothesis must remain `UNRESOLVED`, not PASS. In v0.6.0 investigate full vortex-line topology extraction, long-time/mesh/core convergence, relative equilibria, Floquet/Krein tests and independently held-out oscillations. No mass, gravity or relativity inference is warranted.

16 Windows reproduction with an actual v1.0.9 installation

```
python -m pip install numpy scipy numba pytest
python -m pytest -q tests
run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd REPORT_ONLY
run_all.cmd REVEAL_IF_ALLOWED
```

The original installed `SST_Falsifier_Framework_v1.0.9` is required for official certification. *FREEZE* once per new instance; *BASIC* runs the finite 3D preflight. The new campaign does not change any v0.4.0 evidence.

References

[1] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[2] C. A. Jones and P. H. Roberts, Motions in a Bose condensate IV, J. Phys. A 15 (1982).

[3] SST_CANON-v0.8.36, Sections 1.5, 3.11, 3.14, Appendix P (2026), research programme.

[4] A069 v0.4.0, blinded Hasimoto and radial-GP analytic preflight (2026).